

Physik II — FS 2024 — Prof. R. Wallny — Sheet 4

Release: 12. March 2024 – Due date and discussion: 21. March 2024

Exercise 1: Understanding vector equations and vector operators

Objective: Practice vector analysis.

Let ψ be a scalar field and $\vec{A}$ a vector field. Show the following identities component by component:

a) $\vec{\nabla} \times \vec{\nabla} \psi = \vec{0}$

b) $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$

c) $\Delta \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \vec{\nabla} \times (\vec{\nabla} \times \vec{A})$

Hint: The Laplace operator applied to a vector field verifies $\Delta \vec{A} = \begin{pmatrix} \Delta A_x \\ \Delta A_y \\ \Delta A_z \end{pmatrix}$

Calculate also

d) the gradient of $f(x, y, z) = \cos(x) \ln(y) e^z$.

e) the divergence and the curl of $\vec{a} = \begin{pmatrix} y^2 z^3 \\ 2xyz^3 \\ 3xy^2 z^2 \end{pmatrix}$.

Exercise 2: Electric dipole

Objective: Potential and electric field

An electric dipole consists of two charges with opposite sign but equal absolute values $q_1 = q = -q_2$ that are separated by a distance d . It is defined by its electric dipole moment

$$\vec{p} = q \cdot \vec{d},$$

that points from the negative to the positive charge.

The potential of such a configuration observed from a large distance (i.e. $r \gg d$) is given by

$$\Phi_{Dipol}(\vec{r}) = \frac{p \cdot \cos(\theta)}{4\pi\epsilon_0 r^2}.$$

a) Compare qualitatively the potential of a dipole with the one from a simple point charge.

b) Calculate the electric field of a dipole.

Hint: Use spherical coordinates. Interpret the single components of the field and state its symmetries.

Exercise 3: E-field and electric potential of a charge distribution

Objective: Electric field and associated potential, Gauss's law.

Given is a uniformly charged sphere with constant volume charge density $\rho(x) = \rho_0$ and radius R .

- Find an expression for the electric field inside and outside of the sphere as a function of the distance r from the centre of the sphere.
- Explain the dependence of E on r in both cases.
- Calculate the electric potential of the uniformly charged sphere.

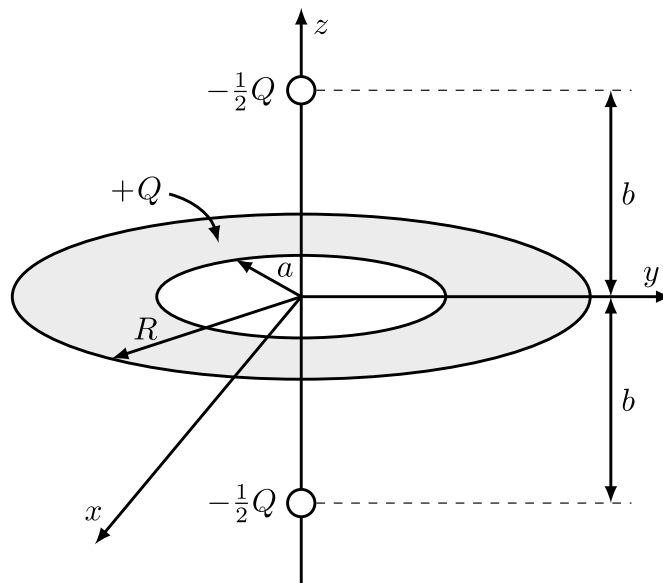
Hint: The potential must be continuous at the surface of the sphere $\Phi_a(R) = \Phi_i(R)$.

- Plot the E-field and the electric potential over the radius.

Hint: Use dimensionless parameters: e.g. normalized radius (r/R).

Exercise 4: Charged circular ring

Objective: Electric potential.



Given is the non-conducting circular disk shown in the figure with radius R and a hole with radius a in the center, where $a < R$. The disk is infinitely thin and has a homogeneously distributed positive charge $+Q$. The coordinate system is right-handed, its origin is located in the centre of the circular disk, and the z -axis runs vertically through the disk. First ignore the two point charges at $\vec{r} = (0, 0, \pm b)$.

- Calculate the electrical potential $\Phi_S(z)$ caused by the disk on the z -axis. Let the potential $\Phi_S(z)$ be zero at infinity: $\Phi_S(\infty) = 0$.

Hint: $\int \frac{x}{\sqrt{x^2 + a^2}} dx = \sqrt{x^2 + a^2} + C$

- b) Calculate the electric field $\vec{E}$ on the z -Achse from the potential $\Phi_S(z)$.

The arrangement is now being extended. A point charge $-\frac{1}{2}Q$ is placed at a distance b above and below the circular disk.

- c) Calculate the electric field $\vec{E}_P(\vec{r})$ created by a point charge $\frac{-Q}{2}$ in its own spherical coordinate system.
Deduce the electrical field created along the z -axis by each of the two point charges.
- d) Calculate the potential $\Phi_P(\vec{r})$ for the two point charges from the electric field $\vec{E}_P$ $\Phi_P(\vec{r})$. Let the potential be zero at infinity ($\Phi_P(\infty) = 0$). Give an expression for the potential created along the z -axis by each of the point charges.
- e) Determine the value a , for which the total potential of disk and point charges vanishes at the origin ($\vec{r} = (0, 0, 0)$): $\Phi_{\text{total}} = 0$, assuming that $R = 2a$.

Exercise 5: Hydrogen atom

Objective: Charge distributions. Electric fields. Gauss' law

A quantum-mechanical treatment of the hydrogen atom shows that the electron in the atom can be treated as a smeared-out distribution of negative charge of the form

$$\rho(r) = -\rho_0 \exp(-2r/a)$$

Here r represents the distance to the center of the nucleus and a represents the Bohr radius, which has a numerical value of 0.0529 nm. The nucleus of a hydrogen atom consists of a single proton which can be seen as a positive point-like charge $+e$.

- a) Calculate ρ_0 , using the fact that the atom is neutral.
- b) Argue why the total electric field of the atom only depends on the radial distance to its nucleus and show that it is given by:

$$E(r) = \frac{ke}{r^2} \exp(-2r/a) \left[1 + \frac{2r}{a} + \frac{2r^2}{a^2} \right]. \quad (1)$$

where k is a constant. Give an expression for k .